

Automated MLB Matchup Analysis and Decision Support Using Pitch-Level Data

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Abstract

Major League Baseball (MLB) generates a vast volume of game, player, and pitch level data vital for matchup strategy and advanced scouting preparation. However, the strategic utility of this data depends heavily on structured collection, systematic validation, contextual content, and readable reports to coaches and players. This project establishes an end to end MLB scouting and game planning system that extracts pitch level data from the official MLB website, processes and validates the information using Python, and stores it within a hosted PostgreSQL database on Railway, this data is transformed into interactive HTML scouting reports. The analytical framework evaluates pitcher arsenals, pitch locations, handedness splits, pitch sequencing, count tendencies, rolling performance averages, contact outcomes, and release characteristics, while separating pitcher performance against left and right handed batters and hitters' performance against left and right handed pitchers. The system processed and validated 1,791 unique player records with complete batting and throwing profiles, reducing errors in matchup specific analyses. The resulting reports organize team, pitcher, hitter, and situational matchup information into an interface designed to optimize pregame preparation and eliminate the manual review of raw pitch level data. This project demonstrates how data engineering, statistical analysis, and interactive visualization can be created into a reproducible decision support workflow with applications for baseball scouting, coaching, and player preparation.

Introduction

Major League Baseball has become increasingly reliant on data driven solutions as advanced player tracking and pitch level data collection have expanded the volume of information available for performance evaluation. A single pitch generates numerous data points, including pitch type, velocity, movement, spin rate, release characteristics, plate location, count, the identity of the pitcher and batter, and outcome. Across an entire season, these observations construct a dataset that offers opportunities to identify patterns in pitcher behavior, hitter performance, and individual matchups. However, while access to detailed baseball metrics has improved significantly, the availability of data does not ensure its usage for strategic decision making. Coaches, scouts, players, and analysts still require to transform thousands of individual data points into actionable insights for in game preparation.

Traditional baseball statistics provide useful summaries of past performance. For instance, a pitcher's metrics may fail to reflect performance discrepancies against left handed versus right handed batters. Similarly, a hitter's overall batting average may conceal performance variations depending on the handedness of the opposing pitcher. Furthermore, pitch selection dynamically shifts based on variables such as the count, batter handedness, pitch sequencing, and the broader game situation.

This operational challenge encompasses both analytical and data engineering problems. Before meaningful matchup statistics can be calculated, comprehensive data spanning games, pitches, players, teams, and rosters must be collected, organized, validated, and aligned. Incomplete player metadata or misidentifications directly compromise downstream analyses. For example, missing information regarding a player's batting side or throwing hand can cause pitch records to be excluded from handedness specific calculations. An advanced scouting system demands more than computational formulas, it requires a data pipeline and a database architecture capable of supporting those calculations.

The primary objective of this project is to develop an MLB decision support system designed to optimize scouting and matchup strategies. The system extracts pitch level data from

official MLB data services, processes and validates the information using Python, stores the verified records within a PostgreSQL database, conducts matchup oriented statistical analyses, and generates interactive HTML scouting reports. Rather than requiring users to manually review raw pitch records or consult multiple sources, the final report synthesizes team, pitcher, hitter, and situational matchup information into navigable sections tailored for pregame analysis.

The analytical framework focuses on converting granular pitch level observations into interpretable tactical tendencies. Pitchers are evaluated using metrics such as pitch usage, velocity, spin rate, whiff rate, called strike plus whiff rate (CSW%), zone rate, chase rate, strikeout rates, walk rates, pitch location, release characteristics, count tendencies, and pitch sequencing. Hitters are assessed based on batting average, strikeout rates, walk rates, swing and miss profiles, pitch type performance, strike zone discipline, and handedness specific results. These performance metrics by pitcher and batter handedness allows the system to overcome season averages and isolate how player tendencies shifts under specific matchup conditions.

Importantly, this system is engineered to serve as a decision support system rather than an automated replacement for human baseball judgment. Statistical tendencies underlie historical patterns within the available data but cannot definitively guarantee the outcome of a future plate appearance or game. Furthermore, coaching and strategic decisions frequently rely on qualitative and contextual variables that are not fully captured within a database, including player injuries, mechanical adjustments, fatigue, weather conditions, opponent strategies, and qualitative scouting observations. The objective is therefore to structure relevant quantitative evidence to complement, rather than supplant, the expertise of coaches, scouts, analysts, and players.

Research Questions

Main Research Question

How can pitch level MLB data be transformed into an automated, reproducible and interpretable decision support system for opponent scouting and match preparations?

Secondary Questions

- To what extent do pitcher behaviors and situational outcomes diverge when contrasting appearances against left handed versus right handed hitters?
- In what ways does hitter efficacy fluctuate when facing opposite handed versus same handed pitching counterparts?
- How can disparate variables including pitch classification, plate location, count leverage, sequencing patterns, release profiles, and rolling averages be synthesized to isolate tactical matchup advantages?
- What architectural requirements must a data pipeline satisfy to ensure the systematic validation and alignment of player, team, and game level metadata for high integrity scouting analytics?
- How can multifaceted pitch level metrics be effectively distilled into an interactive visualization framework that supports actionable pregame strategic planning?

Impact

The primary objective of this project is to synthesize expansive volumes of granular baseball metrics into a structured analytical framework that facilitates strategic game preparation and advanced player evaluation. While detailed pitch level data is increasingly accessible, its utility is fundamentally reliant upon the efficiency with which it can be translated into actionable

intelligence. Coaches, analysts, scouts, and players frequently must evaluate opposing teams within constrained windows, rendering the manual examination of extensive pitches or the querying of multiple datasets highly impractical. The decision support system developed mitigates these operational inefficiencies by integrating systematic data collection, automated validation, statistical refinement, and interactive visualization into a unified scouting workflow.

Stakeholders

Coaches and Baseball Operations Staff

Coaches and baseball operations personnel serve as the primary decision making stakeholders for the system. Prior to a game, the coaching staff may require insights into an opposing starting pitcher's pitch usage, how these tendencies vary by batter handedness or count, and the specific areas of the strike zone most frequently targeted. The system consolidates these observations into a structured interface that supports the formulation of strategic game plans without requiring direct interaction with the underlying database. Furthermore, the system assists in identifying how an opposing lineup performs against various pitch types, locations, and handedness groups. This information supports strategic decisions regarding which components of a pitcher's arsenal offer favorable matchups by providing data on the opponent's pitch metrics, location tendencies, sequencing patterns, and performance against left and right handed hitters. These analyses enable the coaching staff to develop targeted approaches tailored to individual players.

Players

Beyond operational staff, pitchers and hitters represent critical consumers for the analytical output. A pitcher tasked with navigating an opposing lineup can leverage the generated report to discern hitter specific vulnerabilities and strengths, subsequently evaluating these historical patterns with their own unique pitch arsenal. By prioritizing the dynamic interaction between a specific pitcher and their anticipated opponents rather than relying on generic league wide summaries, the system provides a more granular foundation for individualized game strategy.

Hitters utilize these reports to refine their approach against an opposing pitcher. Comprehensive metrics regarding pitch usage, velocity profiles, location clusters, count based tendencies, and sequencing patterns establish a data driven expectation for how an opponent might navigate a given plate appearance. The integration of handedness specific analysis is especially vital, as pitcher strategies frequently exhibit substantial tactical shifts upon whether the hitter occupies the left or right side of the batter's box.

While the system does not seek to dictate every competitive decision, it offers a quantitative evidentiary base. This framework is designed to be synthesized with video review, qualitative scouting insights, personal competitive experience, and real-time in-game adjustments by both players and coaching personnel.

Related Work and Background

The historical trajectory of baseball statistics has transitioned from basic evaluative markers to an expansive analytical landscape that far exceeds traditional metrics such as batting average or earned run average. Sabermetric principles introduced a more rigorous methodology for assessing player value, challenging conventional wisdom and establishing new frameworks for quantifying offensive and defensive contributions. The broad implementation of these

practices within Major League Baseball underscores the pivotal role of quantitative evidence in guiding roster construction, player development, and tactical game management (Lewis, 2003; Tango, Lichtman, & Dolphin, 2007).

This analytical shift is foundational to the current system, as aggregate season long figures often fail to capture the nuances required for specific matchup preparation. For instance, a hitter's overall profile may obscure significant performance variance contingent upon the pitcher's handedness, while a pitcher's summary data might merge divergent tactical approaches used against left versus right handed opponents. By centering the platoon advantage, a strategic concept utilized for over a century, this system leverages the inherent performance fluctuations between batter and pitcher handedness as a primary analytical pillar.

The use of sophisticated pitch tracking technologies has fundamentally altered the precision and volume of available performance data. Since the 2008 implementation of league wide automated tracking, the industry has gained access to standardized velocity and classification metrics. This capability was further enhanced by the 2015 introduction of Statcast, which integrated comprehensive player and batted-ball tracking across every stadium, eventually transitioning to high speed camera systems in 2020. These advancements allow for the examination of granular variables including spin rates, vertical and horizontal movement, extension, and release points that traditional box scores cannot represent.

Platforms such as Baseball Savant provide public access to these metrics, offering visualizations and pitch level search capabilities that inform contemporary scouting. The design of this system moves beyond outcome based evaluation to incorporate the mechanics of how those results are generated. By synthesizing pitch usage, location data, and discipline metrics like chase and whiff rates, the framework constructs a more comprehensive representation of player behavior.

Handedness remains a cornerstone of tactical consideration, primarily through the platoon effect which suggests hitters maintain a distinct advantage when facing opposite handed pitchers. This historical research provides the scaffolding for the scouting reports, necessitating that pitcher and hitter performance be isolated by handedness. Such a framework is particularly critical for managing switch hitters, whose profile is entirely dependent on the throwing hand of the opposition.

By segmenting data the system transforms season averages to reveal how strategies shift under specific conditions. Differences in pitch selection, location tendencies, and discipline behavior are examined through the lens of the specific matchup context, providing a more precise foundation for strategic preparation.

Beyond individual pitch metrics, the sequence and context of a plate appearance are vital components of pitching strategy. The efficacy of a specific pitch is often reliant on deliveries, the current count, and established location patterns. Current research utilizes machine learning and complex statistical models to explore these non random sequential relationships, investigating how setup pitches influence ultimate outcomes.

While these patterns offer structured insights into pitcher tendencies, they do not provide a guaranteed roadmap for future events. In alignment with these findings, this project integrates common sequencing patterns and count based tendencies as scouting indicators rather than predictive absolutes. The goal is to highlight recurring behavioral signatures that inform player preparation while acknowledging the inherent variability of in game execution.

The utility of sports analytics is fundamentally tied to its integration within organizational decision making processes. As emphasized by Alamar (2013), technical sophistication is of

limited practical value if the resulting insights are not interpretable by coaches and players. This philosophy guided the final design of the capstone project, ensuring that analytical outputs are not just raw calculations but are presented through an interactive and accessible interface.

The resulting reports synthesize complex data including strike zone heat maps, release profiles, and matchup recommendations into a unified layer that bridges the gap between data engineering and on field strategy. By transforming granular observations into actionable intelligence, the system serves as a functional tool for those tasked with making high stakes baseball decisions.

While existing literature validates the importance of tracking data, platoon advantages, and sequencing, this project contributes an integrated, end to end workflow for modern scouting. Rather than developing a singular predictive model, the focus is on the synthesis of data acquisition, systematic validation, relational storage, and automated reporting.

Data Engineering

The efficacy of the scouting system is reliant on a robust data engineering pipeline designed to aggregate, structure, and validate baseball metrics prior to statistical refinement. This project prioritizes the construction of a dynamic data architecture over the utilization of a static dataset. The workflow extracts information from Major League Baseball data services, utilizes Python for transformation and processing, and maintains verified records within a hosted PostgreSQL database. This relational structure ensures that the warehouse can support diverse pitcher and hitter matchups across various teams and seasons without necessitating the reconstruction of the core dataset for every analytical request.

Data Sources

The system leverages Major League Baseball data services to procure game, player, and pitch level metrics. Because the necessary attributes for advanced scouting are distributed across multiple endpoints, a multi faceted extraction approach is required.

The MLB Schedule API establishes the foundational game level records, including unique identifiers, dates, and organizational participation. These primary keys are essential for aligning scheduled events with the granular performance data acquired later in the pipeline. The MLB Live Game Feed serves as the primary analytical engine, providing pitch by pitch observations such as velocity, spin rate, movement, plate location, and situational context. By maintaining these records at the lowest level of granularity, the system can calculate performance metrics dynamically rather than relying on aggregate season summaries.

Furthermore, the MLB Player/People endpoints contain the metadata required to bridge numerical identifiers with descriptive profiles, specifically regarding batting side and throwing hand. These attributes are critical for the handedness specific calculations that define the analytical framework. Finally, the MLB Team and Roster endpoints provide organizational context, utilizing expanded roster types to ensure that players who are injured or recently reassigned remain represented within the data warehouse. These integrated sources allow the system to connect every individual pitch to its corresponding game, player identity, and outcome.

Data Pipeline Architecture

Python serves as the primary environment for collection and standardization, ensuring that the records are prepared for storage which is run everyday at 4am pacific to gather the previous games data. By the separation of data acquisition from the reporting layer, the system

allows information to be harvested once and reused across different analyses. PostgreSQL functions as the central data warehouse, hosted via Railway to provide a persistent, cloud based infrastructure that supports remote analytical queries. When a matchup report is initiated, the workflow extracts relevant historical observations and applies statistical formulas to generate pitcher and hitter summaries, which are then rendered into the final HTML interface.

Relational Data Organization

The implementation of a relational database architecture was necessitated by the requirement to preserve high integrity associations between organizational entities, individual athletes, and specific game events. Within this framework, the pitch table functions as the most granular layer, capturing discrete observations rather than relying on static, pre-calculated summaries. This architectural approach enables a dynamic analytical environment where filtering can be applied across dimensions such as seasonal ranges, platoon splits, count leverage, and specific pitch classifications. By deriving metrics from this comprehensive warehouse, the system maintains the flexibility to address evolving matchup scenarios without the need for additional data acquisition cycles. This structural design ensures that the system remains both robust and adaptable, establishing a consistent evidentiary foundation for multifaceted scouting requirements. The following primary tables constitute the core database schema:

Relational Data Organization

The PostgreSQL warehouse is organized into several key relational tables designed to optimize data retrieval and maintain referential integrity across the scouting pipeline:

- **Teams:** Manages organizational identifiers and essential team-level metadata.
- **Players:** Stores critical athlete profiles, including primary positions, batting side, throwing hand, and current team affiliations.
- **Games:** Provides situational context by documenting dates, venue details, participating organizations, and final outcomes.
- **Pitches:** Serves as the central fact table, containing granular pitch-by-pitch observations that link specific pitchers and batters to game events.

Player Metadata and Handedness Validation

The integrity of the scouting reports is heavily dependent upon the accuracy of player metadata, particularly regarding the batting and throwing profiles used for platoon advantage analysis. Missing or incorrect values in these fields directly compromise handedness specific metrics, even when the underlying pitch data is valid. To mitigate this risk, a dedicated validation process was engineered to cross reference roster records with every unique identifier present in the pitch warehouse.

In the final refresh cycle, the system processed 1,791 unique player records identified across the roster collection and pitch observations. Following the systematic retrieval of missing metadata, the system achieved zero omissions for batting side and throwing hand values. This validation effort underscores the critical relationship between data quality and the reliability of the resulting analytical splits.

Team Referential Integrity

Player team relationships require addressing organizational identifiers from external sources that were not represented within the Major League team dimension. The PostgreSQL schema enforces strict referential integrity, which initially caused update failures when

encountering minor league or international organization IDs. To resolve this, the refresh workflow was adjusted to validate team identifiers prior to insertion, clearing 708 non-MLB references while preserving essential player metrics. This strategy ensures the maintenance of a high integrity relational schema while retaining vital analytical information.

Schema Compatibility and Data Validation

Ensuring compatibility between external data representations and internal database constraints was necessary. The pipeline inspects column lengths before insertion to prevent transaction failures caused by descriptive fields that exceed designed limits. Furthermore, data is processed in batches to enhance debugging capabilities and isolate specific record anomalies. This rigorous validation framework ensures that large scale data loads do not obscure underlying quality issues.

Database Reliability

Given the cloud hosted nature of the PostgreSQL warehouse, managing network connectivity and infrastructure configurations was a vital consideration. The project implemented connection verification protocols to ensure stability during complex analytical queries and differentiated between internal Railway networking and public endpoints for local development. These technical considerations are essential for maintaining a reliable automated system capable of consistent data access.

Data Transformation for Analysis

Following validation, the analytical layer transforms granular pitch observations into features by filtering the warehouse for specific matchup variables, such as:

- Individual Pitcher and Batter profiles
- Organizational and Team performance
- Temporal subsets and seasonal ranges
- Pitch classification and throwing/batting handedness
- Situational context including count, inning, and plate location

These subsets facilitate the calculation of advanced metrics like CSW%, whiff rates, and sequencing tendencies, demonstrating the inherent flexibility of the project's data architecture.

Reproducibility and Automation

The pipeline is engineered for reproducibility, allowing scouting reports to be regenerated as the underlying warehouse is updated. By automating the transition from database query to HTML rendering, the system eliminates manual data entry and ensures that analytical outputs remain tied to the current state of the source code and database. Version control via GitHub and managed infrastructure through Railway further support this reproducible decision making workflow.

Data Engineering Summary

The integrated architecture established in this project serves as the scaffolding for advanced player evaluation. The systematic collection, transformation, and validation of pitch level metrics transform raw observations into actionable scouting intelligence. By addressing challenges in metadata accuracy, referential integrity, and infrastructure reliability, the project

creates a stable foundation for matchup specific analysis that complements on-field strategic expertise.

Ethics and Responsible Use

Constructing a data driven scouting framework requires careful ethical consideration regarding data integrity, interpretative transparency, and the responsible application of analytics within athlete evaluation. While this project leverages public performance metrics rather than private personal information, maintaining ethical standards is required for analytical conclusions to significantly impact player assessment and strategic deployment. The system is engineered as a decision support workflow, designed to augment rather than automate professional baseball judgment.

Data Privacy and Appropriate Use

The system utilizes professional baseball metrics acquired via official Major League Baseball data services, focusing exclusively on on field activity such as pitch observations, player identities, and performance outcomes. No sensitive or non competitive personal data is targeted or processed. Furthermore, infrastructure security is prioritized by sequestering database credentials and authentication strings within environment variables, ensuring the codebase remains reproducible without compromising the integrity of the production environment.

Data Quality and Analytical Reliability

The foundational reliability of the scouting reports is fundamentally tied to the precision of the underlying data warehouse. Misaligned player records or incomplete pitch observations can generate skewed statistical conclusions, rendering data validation a core analytical pillar. This was particularly evident when refining platoon splits, missing handedness metadata threatened to exclude valid pitches from specific matchup calculations. By expanding the player refresh cycle to encompass all unique identifiers within the pitch warehouse, the system successfully verified 1,791 player profiles with total metadata completion.

Statistical Uncertainty and Sample Size

Navigating the inherent volatility of baseball performance requires acknowledging the limitations imposed by sample size. As data is segmented into granular situational buckets such as specific counts or pitch-type matchups the volume of observations inevitably diminishes. High whiff rates or strikeout percentages derived from sparse data should be interpreted as indicators rather than absolute capabilities. The generated reports are designed to highlight recurring behavioral signatures while acknowledging that historical patterns do not provide a guarantee for future outcomes.

Context and Potential Bias

Quantitative pitch metrics cannot encapsulate every variable influencing on field results, such as mechanical adjustments, player fatigue, or environmental factors. A framework reliant solely on historical data may inadvertently overweight past behavior when a player has implemented recent strategic shifts. While rolling averages and temporal filtering mitigate these risks, the analytical output should be synthesized with qualitative scouting, video review, and professional expertise to ensure a comprehensive evaluation of an athlete's current profile.

Human Oversight and Decision Support

The scouting reports serve to empower human decision making by structuring relevant evidence for competitive preparation. Observations regarding pitch usage, location clusters, and sequencing patterns establish a data-driven foundation for strategy, yet they do not supersede the necessity of in-game adjustments. Players and coaches must integrate these historical tendencies with real time variables, such as game situation and batter response, maintaining ultimate responsibility for the final execution of tactical plans.

Transparency and Reproducibility

Strategic transparency is maintained by utilizing interpretable baseball metrics rather than predictive models. By documenting the entire data pipeline from extraction to visualization through version control, the project ensures the methodology remains open for examination and reproduction. This approach allows stakeholders to evaluate the observable evidence supporting each analysis, fostering trust in the system as a functional tool for high-stakes baseball operations.

Methodology and Evaluation

The analytical framework was created to synthesize granular pitch level observations into interpretable profiles for pitchers, hitters, and team matchups. Rather than prioritizing a singular metric, the system integrates multiple variables including pitch characteristics, plate location, situational outcomes, handedness, count leverage, sequencing patterns, and rolling performance averages. This comprehensive dataset is changed into an interactive scouting interface designed to facilitate advanced pregame strategic preparation.

Pitcher Analysis

The evaluative process for pitchers begins with the extraction of historical pitch records corresponding to the selected athlete and temporal range. The system gathers these observations to construct both a performance profile and a detailed assessment of individual pitch classifications.

Primary evaluative markers encompass pitch usage, velocity profiles, spin rates, strike efficiency, whiff rates, called strike plus whiff rate (CSW%), zone rates, chase percentages, strikeout and walk rates, batting average against, and contact quality metrics where sufficient data volume permits.

Pitch usage is derived as the frequency of each delivery type within a pitcher's total arsenal, enabling the system to isolate primary weapons from secondary or occasional offerings. Metrics regarding velocity, movement, and plate location are independently calculated for each specific pitch type.

Swing and miss behavior is quantified through whiff rates, while the integration of called and swinging strikes via CSW% offers a more robust measure of strike generation efficacy. Furthermore, zone and chase rates offer insights by differentiating between intentional strikes and pitches designed to elicit expansion of the strike zone.

These indicators are evaluated through a synthesized lens, for instance, a pitch exhibiting a reduced zone rate may maintain high strategic utility if it consistently generates higher chase or whiff rates.

Handedness Specific Pitcher Analysis

Aggregate pitcher statistics frequently conceal critical shifts in approach. Consequently, the system generates distinct performance profiles based on the handedness of the opposing batter. This segmentation allows the scouting report to isolate recurring behavioral including:

- Pitch selection and usage
- Velocity and movement profiles
- Strikeout and walk percentages
- Opponent batting average
- Whiff rates and CSW% efficiency
- Zone discipline and chase behavior
- Plate location tendencies

This filtering process ensures that the handedness specific data represents a precise subset of historical observations rather than a generic summary. Player metadata regarding batting side and throwing hand is sourced from the validated database to maintain analytical integrity.

Batter Analysis

Hitter evaluation employs a parallel framework from an offensive perspective, leveraging pitch level data to assess performance outcomes, swing discipline, and location specific vulnerabilities. Essential metrics include batting average, strikeout and walk rates, swing and miss profiles, and efficacy against various pitch classifications.

Similar to pitcher evaluation, hitters are analyzed through platoon splits to uncover performance fluctuations against left and right handed pitching. Switch hitters receive specialized handling; their effective profile is dynamically interpreted based on the throwing hand of the anticipated opponent.

Strike Zone and Pitch Location Analysis

Plate coordinate data is utilized to map every pitch observation relative to the strike zone. The scouting reports visualize these horizontal and vertical intersections to identify where a pitcher establishes their presence and where a hitter experiences recurring outcomes.

By generating pitch type specific heatmaps, the system isolates how different components of an arsenal attack distinct areas of the zone. These visualizations serve as a descriptive tool to communicate spatial tendencies rather than predicting the exact placement of future deliveries.

Release Characteristics

Observations of the pitcher's release point provide additional context regarding delivery consistency and mechanical profiles. These characteristics are summarized according to the throwing hand to ensure standardized visual representations. This data helps identify if an arsenal originates from a uniform release window, serving as a scouting indicator for pitcher deceptive qualities.

Count Tendencies

Recognizing that pitch selection often shifts according to the ball strike count, the system analyzes historical usage patterns within specific situational states. This highlights when a pitcher might exhibit more predictable behaviors, such as relying on fastballs when behind or increasing breaking ball frequency when ahead in the count.

Pitch Sequencing

Pitches are organized into sequential transitions within each plate appearance to examine recurring patterns. Analyzing common combinations offers deeper strategic context, though the system treats these sequences as historical signatures rather than deterministic roadmaps for future execution.

Inning and Rolling Performance

Temporal trends are examined through inning level tendencies and rolling performance metrics. While season long averages may respond slowly to mechanical or strategic adjustments, rolling measures capture recent fluctuations in player efficacy. These are presented alongside broader data to balance short term sensitivity with long term stability.

Matchup Analysis and Recommendations

The final analytical phase integrates data from opposing pitchers and lineups to identify critical interactions. For offensive preparation, the system isolates likely pitch types and location targets based on the opponent's tendencies. For pitching strategy, hitter vulnerabilities are cross referenced with a pitcher's actual arsenal to suggest viable tactical approaches. These recommendations serve as a quantitative base for strategic planning.

System Evaluation

Evaluation focused on the dual dimensions of analytical correctness and system reliability. Efficacy was verified by comparing generated scouting metrics with the underlying pitch level observations, with particular emphasis on high integrity platoon splits and validated metadata.

Key validation procedures included auditing player identifiers against MLB metadata, verifying pitch type counts, and ensuring that handedness specific calculations referenced the appropriate data subsets. A significant achievement involved the expansion of the player refresh cycle, resulting in 1,791 unique records with zero metadata omissions for batting and throwing side.

Operational refinements to the reporting logic addressed initial discrepancies in displaying handedness summaries, ensuring that each interface page was populated with directly calculated results. This underscores the necessity of aligning statistical validation with the interactive presentation layer.

Results and Visualizations

The execution of this project is a comprehensive MLB scouting and matchup analytical workflow that integrates automated collection, relational storage, statistical refinement, and interactive reporting. The central output consists of dynamic HTML scouting reports that synthesize granular pitch level observations into accessible team, pitcher, and hitter profiles. This architecture allows stakeholders to progress from organizational overviews to specific player tactical signatures without the direct interaction with the underlying PostgreSQL database.

During the development and validation of the reporting engine, a specific matchup between the Los Angeles Dodgers and Detroit Tigers served as the primary testing environment. The resulting report featured Yoshinobu Yamamoto and Tarik Skubal as the starting pitchers, alongside their anticipated starting lineups. This simulation provided a practical assessment of the system's capacity for player identification, handedness validation, individualized performance analysis, and the automated rendering of complex visualizations.

Data Pipeline Results

An achievement of the project was the establishment of a reproducible data architecture rather than a static analytical snapshot. The pipeline enables the systematic harvesting of game, pitch, roster, and athlete metadata, which is then structured within a persistent warehouse for subsequent retrieval. This separation of acquisition and analysis ensures the system remains scalable across various teams and seasons.

The validation of player metadata proved fundamental, as the accuracy of the platoon advantage calculations relies entirely on the integrity of batting and throwing profiles.

This validation secured the reliability of the handedness specific analytical segments, demonstrating that high integrity scouting intelligence is fundamentally dependent on the quality of the underlying dimensional data.

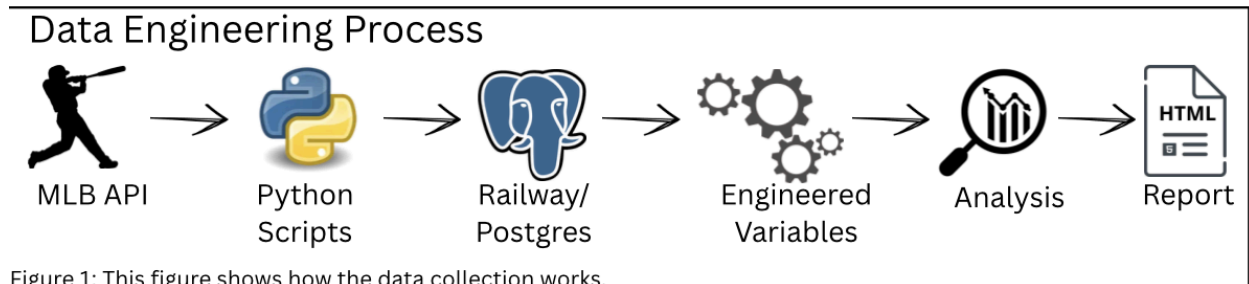


Figure 1: This figure shows how the data collection works.

Team Level Matchup Overview

The initial analytical layer establishes organizational context by evaluating team level offensive performance and situational trends. Rather than immediately presenting granular pitch data, the system prioritizes broader metrics that characterize the offensive environment of each participant.

The integration of rolling averages allows for the identification of recent performance fluctuations, which may diverge from season long expectations. By the usage of short term trends with aggregate metrics, the system provides a more nuanced understanding of an opponent's current efficiency. This overview serves as the foundation for the scouting workflow before the user explores more specific player profiles.

[Insert Figure 2: Team Matchup Overview / Rolling Performance] Need to add it

Figure 2. Example team level matchup overview demonstrating offensive trends and recent performance.

Pitcher Profile and Arsenal

The pitcher analysis begins with a summary of the athlete's overall profile and a detailed classification of their current arsenal. The report identifies recurring pitch types, quantifying the frequency and physical characteristics of each individual delivery.

Analyzing pitch usage is critical for establishing a data driven expectation for hitters. The arsenal view effectively separates primary weapons from occasional secondary offerings, while metrics including velocity, spin rate, whiff rate, CSW%, and zone discipline provide an evaluation of effectiveness. This synthesis allows the user to interpret how a pitcher navigates the strike zone and how hitters respond to specific components of the arsenal. The system utilizes

compact visual summaries rather than exhaustive tabular data, facilitating the rapid identification of a pitcher's tactical signature.

[Insert Figure 3: Pitch Arsenal Visualization]

Figure 3. Pitch arsenal visualization illustrating the frequency and metrics of primary delivery types.

Handedness Specific Results

A core analytical pillar of the final system is the isolation of performance metrics by the handedness of the opposing batter. This segmentation overcomes the limitations of aggregate season averages by generating independent profiles for appearances against left and right handed hitters.

Statistical variables such as batting average allowed, strikeout rates, and pitch usage patterns are compared across these matchup conditions to reveal tactical shifts. This distinction is vital, as pitchers frequently alter both their selection and location tendencies based on the hitter's side of the box. The validation of player metadata ensures that these platoon splits are derived from a complete evidentiary base, minimizing the risk of analytical exclusion.

[Insert Figure 4: Overall vs. LHB vs. RHB Comparison]

Figure 4. Comparison of performance profiles across opposing batter handedness. Strike-Zone and Pitch-Location Visualizations

Spatial analysis of plate coordinate data enables the visualization of pitch location patterns relative to the strike zone. By mapping horizontal and vertical intersections, the system identifies where a pitcher establishes their presence and where hitters exhibit recurring outcome signatures.

The reports provide both aggregate location heatmaps and specialized views for individual pitch types. This granularity is essential for distinguishing the different strategic zones targeted by various deliveries within a pitcher's arsenal. To ensure visual consistency and ease of interpretation, the system maintains a standardized strike zone framework for both pitcher and batter analysis.

[Insert Figure 5: Pitch-Type Strike-Zone Visualization]

Figure 5. Spatial distribution of a selected pitch type relative to the strike zone. Release-Point Visualization

To provide additional context regarding delivery consistency, the system includes release point visualizations that reflect the pitcher's unique mechanical profile. These observations are standardized according to the throwing hand to ensure a coherent visual orientation for all athletes.

The offensive scouting architecture parallels the pitcher framework, utilizing pitch level data to evaluate hitter efficacy, discipline, and spatial vulnerabilities. Individual reports synthesize performance markers, platoon splits, and location specific outcomes into a unified offensive profile.

By segmenting results by pitcher handedness, the system reveals performance fluctuations that are often masked by overall statistics. Detailed metrics regarding swing behavior and pitch type performance are presented alongside location visualizations, while the

consistent design language across the interface minimizes the cognitive load required for navigation.

[Insert Figure 7: Batter Handedness / Strike-Zone Analysis]

Figure 7. Example hitter profile integrating handedness specific analysis and location patterns.

Matchup-Oriented Scouting Output

The final analytical phase integrates insights from both opposing pitchers and lineups to support strategic pregame preparation. Rather than presenting static box score data, the reports are organized around the dynamic interaction between the specific athletes participating in the matchup.

For offensive strategy, the report identifies pitcher tendencies that should influence the hitter's approach, including usage patterns and sequencing signatures. For pitching preparation, hitter vulnerabilities are cross referenced with the available arsenal to identify tactical advantages. This focus ensures that the analytical output serves as a functional scouting tool tailored to the immediate competitive context.

Interactive Report Organization

The usability of the reports was enhanced through an iterative design process that prioritized navigation and visual clarity. The final HTML interface utilizes a tabbed structure to separate team overviews, pitcher analytics, and hitter scouting, eliminating the need for excessive scrolling through a single continuous document.

This organization allows users to efficiently transition between various analytical depths. Furthermore, visual clutter was reduced by removing redundant labels and low volume data fields that did not contribute to strategic decision making. These refinements ensure that the metrics and visualizations most relevant to scouting remain the primary focus of the interface.

Discussion, Limitations, and Future Work

Discussion

This project underscores the fundamental obstacle in constructing a strategic scouting framework. A functional analytical system is reliant upon systematic data collection, robust database architecture, rigorous validation, and an interface that distills complex results into interpretable intelligence. This platform integrates these disparate components into a unified scouting workflow, facilitating the transition from granular pitch observations to actionable team, pitcher, and situational profiles.

An observation during development involved the direct correlation between the integrity of the data warehouse and the reliability of resulting analytics. Platoon specific segments initially revealed inconsistencies stemming from incomplete athlete metadata. Since handedness based splits require the identification of batting and throwing profiles, missing records compromise downstream calculations. By expanding the refresh cycle to include every unique identifier within the pitch warehouse, the system successfully verified 1,791 player records. This effort demonstrates how refinements to the data engineering layer fundamentally enhance the evidentiary base of the final scouting product.

Furthermore, the development cycle emphasized the necessity of matchup context in player evaluation. While aggregate season long figures provide a baseline, they often obscure

significant tactical shifts. Pitchers frequently alter their usage patterns, location tendencies, and strikeout efficiency when contrasting appearances against left versus right handed opponents. Segmenting these performance profiles allows the system to overcome generic averages and isolate recurring behavioral signatures tailored to specific matchup conditions.

The project also highlights the strategic utility of synthesizing statistical summaries with interactive visualizations. While metrics such as whiff rate, CSW%, and discipline percentages offer numerical refinement, spatial tools including strike zone heatmaps and release point profiles render behavioral patterns more accessible. By combining quantitative evidence with visual indicators, the scouting reports provide a more comprehensive representation of player behavior than a singular analytical approach.

Finally, refinements to the interface demonstrated that excessive information can detract from analytical utility. Earlier versions included redundant metrics and explanatory text that increased cognitive load without contributing to strategic planning. The final tabbed format prioritizes information relevant to pregame preparation, reflecting a core principle of decision support design.

Limitations

Despite the established workflow, several limitations influence the interpretation of the analytical results.

The primary constraint involves sample size, as baseball metrics become increasingly sparse when segmented into granular situational buckets. While a pitcher may possess an expansive total of observations, data regarding specific counts or pitch type matchups against a particular handedness may be limited. Statistics derived from these diminished subsets should be viewed as indicators.

Additionally, the system primarily uses historical tendencies. Past behaviors do not provide a guaranteed roadmap for future execution, especially as players and coaching staff implement mechanical adjustments or strategic shifts in response to identified weaknesses.

The analytical framework does not establish a causal link between scouting recommendations and specific performance outcomes. While the project demonstrates a consistent capacity to validate and present matchup intelligence, it does not quantitatively prove that these suggestions increase the probability of achieving competitive success, such as recording outs or scoring runs.

Contextual variables including player fatigue, injury status, weather conditions, and ballpark effects remain largely outside the current database architecture. These qualitative and environmental factors frequently influence field strategy but are not fully captured within historical pitch level observations.

Infrastructure reliability represents a further operational limitation. Because the PostgreSQL warehouse is cloud hosted, the generation of reports is reliant upon network stability and database availability. Complex analytical queries may occasionally experience transaction failures or increased runtimes due to remote connectivity constraints, highlighting the technical requirements of maintaining a persistent automated system.

Furthermore, the computational load required to process multifaceted pitcher and hitter profiles can affect system efficiency. As the volume of features expands, the time required for HTML rendering increases. Future technical optimizations, such as materialized tables or enhanced query indexing, would be necessary to support larger scale analytical requests.

Finally, the system is currently tailored for starting pitcher and offensive preparation rather than encompassing a complete game strategy engine. Areas such as bullpen analysis, defensive positioning, and late game substitution patterns are less developed. The reports should therefore be utilized as a foundation for pregame scouting rather than an exhaustive tool for all strategic decisions.

Future Work

The established architecture provides the scaffolding for several prospective analytical enhancements. A vital next step involves prospective validation, where scouting recommendations are compared against actual game outcomes. By documenting tendencies prior to competition and evaluating them against real time pitch selection and situational results, the system could provide a more rigorous assessment of the practical predictive value of historical signatures.

Integrating expected runs and win probability frameworks would further connect individual matchups to organizational goals. By estimating how specific tactical choices influence scoring probabilities, the system could transition from descriptive scouting toward a more evaluative game management tool.

The application of Monte Carlo simulation offers the capacity to model thousands of simulated games based on player outcome probabilities. This would facilitate the comparison of alternative lineups and pitching strategies within a probabilistic environment, establishing a more advanced foundation for pregame strategic planning.

Expanded contextual variables, such as pitcher workload, ballpark dimensions, and umpire strike zone tendencies, would provide additional layers of situational nuance. Incorporating these factors would refine the analysis beyond individual pitcher hitter interactions.

Developing a dedicated bullpen scouting component represents a critical opportunity for growth. Applying the existing handedness and arsenal analytical framework to relief options would allow the system to support strategy through every inning of competition.

Finally, engineering enhancements including automated data refreshes and query optimization would improve the scalability of the decision support workflow. These technical refinements would ensure that the system remains both efficient and accessible for daily organizational use.

Discussion Summary

This project demonstrates that meaningful baseball analytics requires the integration of data acquisition, systematic validation, and interactive visualization. The resulting framework provides a reproducible foundation for player evaluation and strategic game preparation by transforming raw metrics into interpretable intelligence.

Ultimately, the system serves as a decision support workflow designed to complement the expertise of professional baseball personnel. While it establishes a stable analytical architecture, it offers clear pathways for future extensions in predictive modeling, simulation, and comprehensive game strategy analysis.

Conclusions and Recommendations

This capstone project of gathering granular Major League Baseball metrics into a reproducible analytical framework for scouting and decision support. By integrating collection, automated validation, relational storage, and interactive reporting into a unified workflow, the system demonstrates that each stage fundamentally dictates the strategic utility of the final output. The project moves beyond independent technical tasks to establish a pipeline where data engineering directly informs the quality of athlete evaluation.

The primary research objective investigated the transformation of pitch level observations into a navigable and interpretable framework for pregame strategic planning. The resulting platform addresses this challenge by utilizing Python for data processing, maintaining a persistent PostgreSQL warehouse, and rendering matchup oriented insights into an interactive HTML interface. This project empowers stakeholders to explore organizational context, pitcher arsenals, hitter profiles, and spatial tendencies without necessitating direct interaction with the underlying database infrastructure.

A pivotal conclusion of this research is that granular matchup context reveals tactical signatures often misled by aggregate season long metrics. Performance fluctuations across handedness, pitch classification, count leverage, and sequencing patterns establish a more precise foundation for understanding competitive outcomes than generic averages. Consequently, advanced scouting workflows should prioritize these specific situational subsets, provided they are supported by a sufficient evidentiary base, to overcome limitations of summary statistics.

Regarding presentation, the development cycle demonstrated that increasing the volume of available metrics does not inherently improve strategic utility. By removing redundant variables and reorganizing the interface into specialized visualizations, the final design prioritizes information that contributes directly to in game preparation. This evolution suggests that effective decision support systems must balance analytical sophistication with visual clarity and usability to bridge the gap between raw data and actionable intelligence.

Ultimately, this project establishes a stable architecture for modern baseball decision support by bridging the gap between data engineering and on field strategy. By transforming granular observations into structured evidence, the system provides a reproducible workflow that empowers coaches, scouts, and players while preserving the vital role of human judgment in the final execution of tactical plans.

Reproducibility

The system is engineered for reproducibility through the integration of version controlled source code, a persistent relational warehouse, and automated scripts for processing and reporting. By maintaining the codebase within a GitHub repository, the development of data acquisition, database management, and analytical modules remains transparent and verifiable throughout the project lifecycle.

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